Design of Integrated LLCT Module for LLC Resonant Converter

Wenduo Liu, J.D. van Wyk
Center for Power Electronics Systems
The Bradley Department of Electrical and Computer Engineering
Virginia Polytechnic Institute and State University
Blacksburg, VA 24061 USA
wdliu@vt.edu, daan@vt.edu

Abstract— This paper presents the electromagnetic design of integrated L-L-C-T modules with stacked structure for an LLC resonant converter. Based on the one-dimensional magnetic field assumption, both the electromagnetic design and the loss calculation are investigated. A high power density design sample is provided and the test result is presented.

I. INTRODUCTION

The trend of pursuing high power density DC/DC converters leads us to develop converters capable of operating at high switching frequencies with high efficiency. For this purpose, an LLC resonant converter has been proposed for Front End DC/DC conversion in Distributed Power System [1]. This converter is designed to operate with high efficiency at high frequencies, up to 1 MHz. A circuit diagram is shown in Fig. 1.

At a high frequency, the volume of the passive components of a converter can be reduced. However, in the state-of-art Front End DC-DC converter, the discrete passive components and the heat sink still occupy more than 80% of the total system volume [2]. To reduce the volume of the magnetic components, magnetic integration has been proposed [3]. By integrating the resonant inductor, the magnetizing inductor and the transformer, the number of magnetic components is reduced. However, the integrated module still has a large profile, and this method is not capable of integrating the resonant capacitor into the module. To further reduce the overall volume and improve the power density, passive integration technology is applied to integrate the resonant inductor and capacitor, the magnetizing inductor, and the transformer (L-L-C-T) into one module.

Instead of the previously proposed planar spiral winding structure [4-6] and symmetric single-layer structure [7], the stacked structure is selected to design high efficiency and high power density samples. In this paper, the theory of passive integration is demonstrated in Section II, and then the stacked structure is introduced. In Section III, the electromagnetic field distribution on the cross-section of the module is investigated with a one-dimensional model; and the electromagnetic design procedure is presented. In Section IV, the power loss is calculated by applying loss models for the calculation of core loss, conduction loss and dielectric loss, respectively. Based on the field distribution model, the conduction loss is decomposed into skin effect loss and proximity effect loss. With the proposed loss model, the factors affecting the conduction loss are investigated. In Section V, the design for a specific example is proposed. The sample is tested and the results are presented.

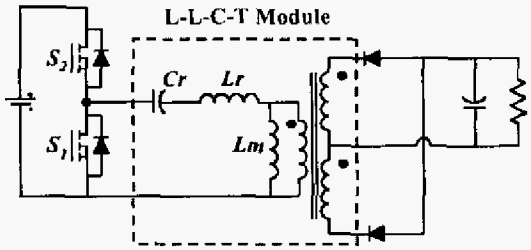


Fig. 1 LLC Resonant Converter with discrete passive components

II. PASSIVE INTEGRATION TECHNOLOGY AND THE STACKED STRUCTURE

A. Integrated cell

The typical structure applied in order to integrate the capacitor and the inductor is a long, thin dielectric substrate covered with two conductor layers, as shown in Fig. 2 (a). This is called an integrated cell. There are four terminals in one cell. When two terminals are arbitrarily selected and excited with AC waveforms, the conductors carry conduction current and the dielectric substrate carries displacement current. The cell stores and transforms both electrical and magnetic energy, and functions as the combination of inductors and capacitors.

For such a transmission structure, the current and voltage distribution along the current flowing direction has been investigated with analytical expressions. When the wavelength of the excitation waveform is much larger than the cell dimensions, both voltage and current distribution along the cell length can be regarded as linear. Fig. 2 (b) shows an example of this.

The impedance and phase versus frequency curves represent the electromagnetic behavior of the cell. Fig. 3 shows the behavior of the cell, when the input waveform is exerted on the terminals *a* and *d*. It is obvious that the cell functions differently in different frequency ranges. At the frequency ranges around the first resonant point, the cell

This work was supported primarily by the ERC Program of the National Science Foundation under Award Number EEC-9731677.

functions as an LC-series resonator; while at the frequency range around the second resonant point, the cell functions as a parallel resonator.

So far it has been reported that the electromagnetic behavior of integrated passives can be accurately predicted at frequencies lower than one hundred mega hertz [8]. Nevertheless, high-accuracy prediction is not necessary for the electromagnetic design. For the passives used to store and transform electromagnetic energy in power conversion circuits, normally, only the behavior around the first resonant frequency of concern.

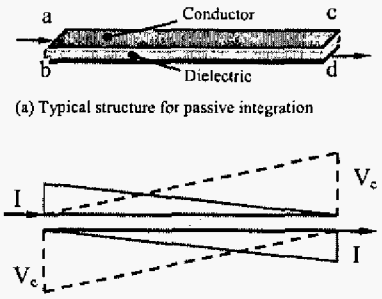
In practical applications, power passives need to transform a large scale of power. Thus, a magnetic core is added to store extra magnetic energy. When the cell is surrounded with a core, the conduction current and dielectric current together induce magnetic flux inside the core, and the magnetic energy is stored in the air gap. To store sufficient electrical energy, usually more than one cell is in the integrated module. The large numbers of cell terminals provide numerous possibilities to make interconnections, and lead to great design flexibility.

B. Stacked Structure

Based on the passive integration theory, a stacked structure is proposed to integrate the LLCT module. As shown in Fig. 1, the basic part of an LLCT module is a transformer with a small magnetizing inductance. The leakage inductance is exploited as the integrated resonant inductance at the primary side. By replacing the conductor winding with internal cells at the primary side, the resonant capacitor can be integrated into the module.

The exploded view of an LLCT module with the stacked structure is shown in Fig. 4. Two sets of planar E core and I core constitute the two winding windows. In each winding window, four internal cells are stacked vertically. The cells are connected at the terminals through the interconnections on the vertical surface at the two ends of the module. The cell dimensions and the properties of the dielectric substrate determine the resonant capacitance. The conduction current and displacement current form the primary side current loop and create magnetic flux inside the core. By changing the thickness of the air gap, the magnetizing inductance can be adjusted to the required value. With an inserted leakage layer between the primary and secondary windings, the leakage inductance is also controllable. The secondary winding is made of thin copper strips. For ease in identification of directions, a coordinate system is also defined in Fig. 4.

The electromagnetic behavior of the primary winding with the core is determined by the applied interconnections [9-10]. The most-frequently applied interconnection is connecting the cells in parallel. Fig. 5 shows the interconnections of an LLCT module with 8 cells in the primary winding and the center-tap structure in the secondary winding. By short-circuiting the secondary winding, the module functions as an LC-series resonator at the first resonant frequency, which is determined by the leakage inductance and the cell capacitance. The second resonant frequency is determined by the leakage inductance and the inter-winding capacitance, and is normally much higher than the first resonant frequency. In order to examine the behavior around the first resonant frequency only, the first order model using lumped parameters, as shown in Fig. 1, can represent the integrated module and is applied in the electromagnetic design and loss calculation.



(a) Typical structure for passive integration

(b) Voltage and current distribution

Fig. 2 Typical structure for passive integration

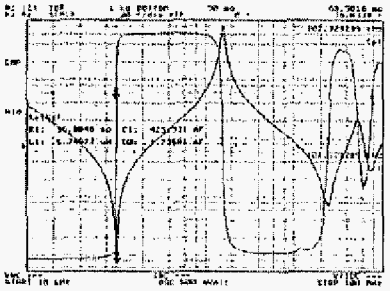


Fig. 3 Impedance and phase vs. frequency – behavior of a typical integrated passive

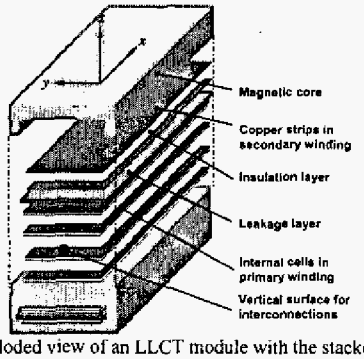


Fig. 4 Exploded view of an LLCT module with the stacked structure

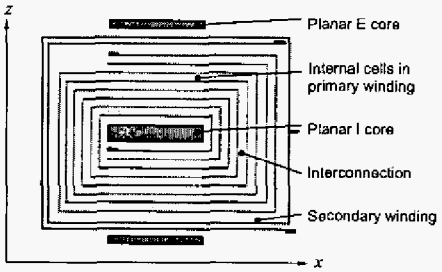


Fig. 5 Interconnections and winding structure of an integrated LLCT module

III. Electromagnetic Design

A. One-dimensional magnetic field

As shown in Fig. 6, the cell width is much larger than the cell height, and the cell width is close to the winding window width. Thus, the magnetic field is considered to be parallel with the cell surface and uniformly distributed along the y-axis. The magnetic field on the yz plane is a function of the single variable z. This means that on the cross-section of the stacked structure, a one-dimensional magnetic field can be assumed.

A 2D simulation was done in the program MAXWELL to investigate the magnetic field distribution. In the simulation, all the eight conduction layers in the primary winding carry alternating current I. The current in the two secondary winding conductors has the amplitude $4I$, and has a 180° phase shift.

With the assumption of a one-dimensional magnetic field, the field distribution on the yz plane can also be estimated with Ampere's law. Since the core has much higher permeability than the air, the magnetic field strength inside the core is negligible, and then the magnetic field is concentrated in the winding window. By assuming that the current density in the winding window is distributed uniformly along the z-axis, the estimated field distribution is a linear function of z. Both the simulation result and estimated result of the field distribution are shown in Fig. 6. For ease in calculation, the estimated field distribution is applied in the electromagnetic design.

B. Electromagnetic design

The magnetizing inductance, L_m, is related to the turn's number in the primary winding, the core dimensions, and the thickness of the air gap in the magnetic flux loop.

The inherent leakage inductance in the primary winding is induced by the flux linking to the primary winding only. The value of the inherent leakage inductance is dependent on the primary winding dimensions. With the known magnetic field distribution, the inherent leakage inductance can be estimated by calculating the magnetic energy stored in the winding window, with the secondary winding short-circuited. The energy stored in the two winding windows is expressed as

$$E_w = 2 \cdot w_{win} l_{core} \left[\int_0^{h_{pwin}} \frac{1}{2} \mu_0 \left(\frac{n_p I_p z}{w_{win} h_{pwin}} \right)^2 dz + \int_0^{h_{swin}} \frac{1}{2} \mu_0 \left(\frac{n_p I_p z}{w_{win} h_{swin}} \right)^2 dz \right] \quad (1)$$

where h_{pwin} and h_{swin} refer to the height of the primary winding and the secondary winding; w_{win} is the winding window width, n_p is the turn's number in the primary winding; I_p is the current in the primary winding.

Thus, the inherent leakage inductance is calculated as

$$L_{\text{int}} = \frac{2E_w}{I_p^2}. \quad (2)$$

In a practical design, the inherent inductance should be lower than the design requirement. A leakage layer is inserted between the primary and secondary windings to provide an additional path for magnetic flux, so that the overall leakage inductance can be adjusted to the desired value.

To have a relatively accurate calculation of the additional leakage inductance, the reluctance of the magnetic core in the flux loop is also taken into account, as shown in

$$L_{add} = 2 \cdot \frac{\mu_0 n_p^2}{\dfrac{2l_{gap}}{A_{core}} + \dfrac{2h_{pri} + 2T_{core} + w_{win}}{\mu_r A_{core}} + \dfrac{w_{win}}{\mu_L A_L}}, \quad (3)$$

where l_{gap} is the air gap height, A_{core} is the core cross-sectional area; T_{core} is the core thickness; μ_r and μ_L are the relative permeability of the core and the leakage layer material respectively.

The capacitance provided by the primary winding is mainly related to the cell dimensions and the dielectric material properties. The capacitance of one cell is

$$C_{cell} = \frac{\varepsilon_0 \varepsilon_r A_{cell}}{h_{di}}, \quad (4)$$

where ε_r is the relative permittivity of the dielectric material; h_{di} is the dielectric substrate thickness; and A_{cell} is the area of the copper on the cell surface. For an n-cell winding with parallel interconnections, as shown in Fig. 5, the overall capacitance is n times of the cell capacitance.

C. Design constraint and optimization

It is obvious that function of copper layers in the cells is two-fold. As conductors, the copper layers provide a path for the conduction current, which generates the magnetic flux inside the core. The length of the copper layers is related to the turn's number of the primary winding, and then both the magnetizing and resonant inductance. In additional to functioning as conductors, the two copper layers of a cell form a capacitor and store electrical energy. Along with the dielectric thickness, the area of the copper layer influences the capacitance. Therefore, the cell dimensions influence both the inductance and the capacitance.

In practical applications, the material characteristics cannot be realized to arbitrary values as required. The result of this is that the specific cell dimensions may not be able to meet the requirements of the inductance design and that of the capacitance design at the same time. Hence, a constructional limitation is introduced into practical designs. A design map is needed to determine the optimal design with the maximum power density.

In the LLCT module design, the magnetic portion is designed first. The core material is selected according to its operating frequency. Then, the turn's number and core cross-sectional area are calculated. According to the primary winding turn's number and the required capacitance, the dielectric material is selected and the cells dimensions are set. Then, the material and dimensions of the leakage layer are determined. Once all the dimensions and material properties are determined, the power loss of each part can be calculated.

To optimize the design, the average current density, J, and the magnetic flux density, B, are set as the two design variables. By changing the values of the variables, a set of designs can be obtained. The leakage inductance, L_r is set as the design criterion to filter out unfeasible designs. Those

designs meeting the design specifications are investigated with thermal simulation for thermal management.

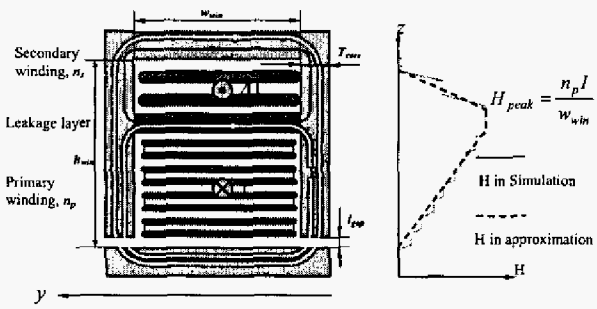


Fig. 6 One-dimensional magnetic field distribution on the cross-section of the top half of the LLCT module

IV. Loss Calculations

The power loss of an integrated LLCT module includes core loss, copper loss and dielectric loss. The lumped parameter model in Fig.1 is applied in the circuit simulation to obtain the voltage and current waveforms, which are used to calculate the losses.

A. Core loss

As shown in Fig. 7, the voltage across the magnetizing inductor is approximately a square waveform. The Generalized Steinmetz Equation (GSE) can be applied to calculate the core loss [11]. The core loss density is estimated as

$$P_v = \frac{1}{T}\int_0^T k_1 \left|\frac{dB}{dt}\right|^{\alpha} \left|B(t)\right|^{\beta-\alpha} dt\,, \tag{5}$$

where α and β are the Steinmetz parameters of the core; k_1 is a coefficient determined by α and β; B is the flux density.

Both the magnetic flux $B(t)$ and the item $\frac{dB}{dt}$ are functions of time. With the data extracted from the voltage and current waveforms, the voltage and current of the magnetizing inductor can be expressed in Fourier series as the functions of time, $V(t)$ and $I(t)$. By applying Ampere's law, the magnetic flux $B(t)$ can be derived from the current function $I(t)$ as

$$B(t) = \frac{n_p}{\frac{l_{gap}}{\mu_0} + \frac{MML}{\mu_0 \mu_r}} \cdot I(t), \tag{6}$$

where MML is the mean magnetic flux length.

By applying Faraday's law, the changing rate of the flux density $\frac{dB}{dt}$ can be derived from the voltage function $V(t)$, as

$$\frac{dB(t)}{dt} = \frac{1}{n_p \cdot A_{core}} \cdot V(t). \tag{7}$$

Due to the application of the leakage layer, the magnetic flux density inside the core is uneven. The variation in flux density, caused mainly by the leakage layer is taken into account by sectionalizing the structure.

The power loss in the leakage layer is difficult to estimate due to the lack of the data of material properties. However, since the volume of the leakage material is small, the loss is neglected.

B. Dielectric loss

When operating at low frequencies around the first resonant point, the voltage between the top and bottom copper layers of a cell is distributed uniformly along the x-axis. Thus, in the dielectric loss calculation, the parallel-connected cells can be regarded as a discrete capacitor. The voltage across the resonant capacitor in the lumped-parameter model is used to calculate the dielectric loss as

$$P_d = \omega C V_{ac-coupled}^2 \tan\delta\,, \tag{8}$$

where $V_{ac-coupled}$ is the rms value of the AC voltage across the resonant capacitor; $tan\delta$ is the loss factor, which is normally within the range of [1%, 3%].

C. Conduction loss

1. Conduction loss in the primary winding

In the primary winding, the current distribution in different copper layers is different. Fig. 8 shows the current distribution along the cell length in an 8-cell module. For each copper layer, the current distribution is an approximately linear function of the distance x, $I(x)$. Correspondingly, the magnetic field strength is also a function of x, $H(x)$.

With the assumption of one-dimensional magnetic field distribution at the yz plane, the orthogonal principle of skin effect and proximity effect can be applied to calculate the skin-effect loss and the proximity-effect loss separately [13]. The skin-effect loss and proximity-effect loss of a copper layer can be calculated with (9) and (10), respectively.

$$P_{skin} = \int_0^l \frac{I^2(x)}{4w\sigma\delta} \cdot \frac{\sinh\nu + \sin\nu}{\cosh\nu - \cos\nu} dx \tag{9}$$

$$P_{prox} = \int_0^l \frac{H_e^2(x) w}{\sigma\delta} \cdot \frac{\sinh\nu - \sin\nu}{\cosh\nu + \cos\nu} dx \tag{10}$$

where l is the cell length; w is the width of the copper layer; σ is copper conductivity; δ is the skin depth of copper at the operating frequency; ν is the ratio of the copper thickness to the skin depth; $I(x)$ is the current distribution with respect to x; and $H_e(x)$ is the equivalent magnetic field distribution.

For each copper layer, the magnetic field on the top surface, $H_{top}(x)$, is different from that on the bottom surface, $H_{bot}(x)$, due to the current flowing in the copper layer. Thus, the copper layer is in a non-uniform magnetic field. As proved by Ferreira in [14], the proximity-effect loss of a conductor in an non-uniform magnetic field is equal to that in a uniform field with the equivalent field strength as

$$H_e(x) = \frac{H_{top}(x) + H_{bot}(x)}{2}. \tag{11}$$

2. Conduction loss in the secondary winding

In the secondary winding, the current distribution is uniform along the winding length. Thus, both the current

distribution and the magnetic field distribution stay constant along the x-axis.

In the center-tap structure, the secondary winding consists of two independent windings. Each winding operates during half of the operation period. In terms of the, the two windings are exactly the same. The skin-effect loss in the two windings is the same and can be calculated with (9). Since each winding operates for only half a period, the skin-effect loss is half of the calculated value.

In the stacked structure, the two copper layers of the secondary winding are in different positions. Thus, the magnetic field exerted on the two copper layers is different. As shown in Fig. 9, in the first half-period, the inner copper layer carries current, while the net current in the outer copper layer is zero. The magneto-motive force on the outer layer is almost zero, and then the magnetic field exerted to the outer copper layer is zero. That means there is no proximity effect loss in the outer layer. In the second half-period, the outer layer carries current. Though the net current in the inner layer is zero, the magnetic field exerted on the inner copper layer reaches the maximum value. In this case, the proximity effect loss in the inner layer is significant.

C. Factors affecting the conduction loss

A specific example was selected to study the factors affecting the conduction loss. It assumed that a stacked structure sample with eight cells in the primary winding and two turns in the secondary winding operates at 1 MHz. The cell width is 10 mm, and the cell length is 60 mm. The copper thickness is equal to the skin depth. The space between two adjacent copper layers in the primary winding is twice the skin depth.

The first seven harmonics of the current waveforms were extracted from the circuit simulation results and applied in the calculation of conduction loss. Fig. 10 shows the percentage that each harmonic contributes to the conduction loss. It shows that in the LLC resonant converter, the conduction loss induced by the fundamental component is dominant.

Fig. 11 shows the conduction loss in each layer. The inner layer of the secondary winding contributes most of the conduction loss, about 22%. The loss in the secondary winding is about 78% of the total conduction loss.

By iterating the loss calculation by varying the copper thickness, the optimum copper thickness for the primary and secondary winding can be found. As shown in Fig. 12, when the copper thickness is about 60% of the skin depth, the conduction loss in both the primary and secondary winding is at its minimum; when the copper thickness is 3 times of the skin depth, the conduction loss in the primary winding is at its maximum. It also indicates that in a one-dimensional magnetic field, increasing the copper thickness will finally reach a constant conduction loss. This is because the increasing copper thickness reduces the DC resistance, but increases the ratio of AC resistance to the DC resistance.

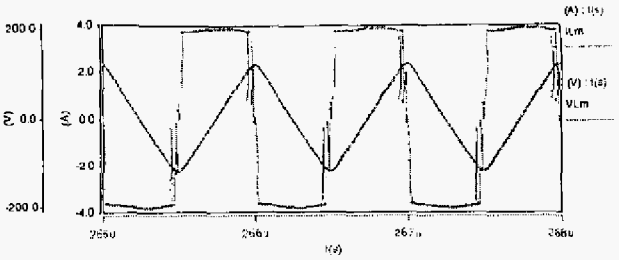


Fig. 7 Voltage and current waveforms of the magnetizing inductor

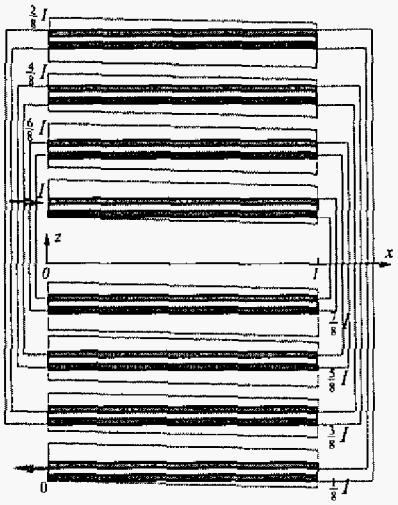


Fig. 8 Current distribution in the primary winding of an 8-cell module

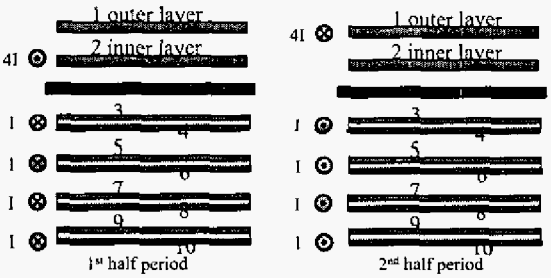


Fig. 9 Current distribution in different half-periods

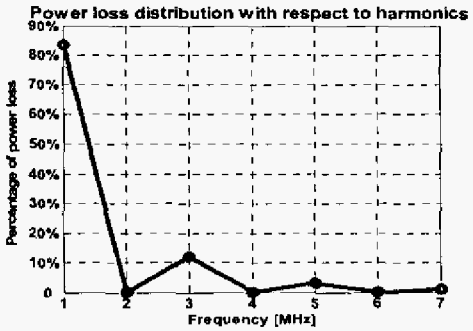


Fig. 10 Percentage of conduction loss induced by harmonics

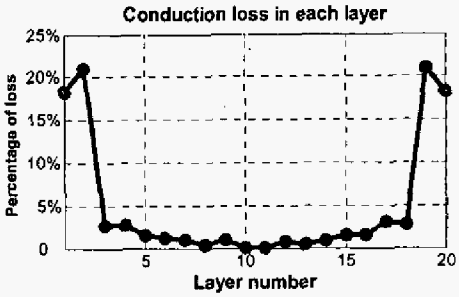


Fig. 11 Conduction loss in each layer

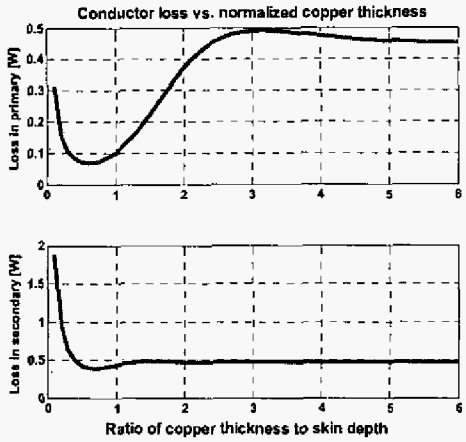


Fig. 12 Conduction loss vs. normalized copper thickness

V. Design Example and Experiment

A design sample was made according to the parameters of a 500W 1 MHz LLC resonant converter. The input voltage of the converter is 200V, the output is 48V, the input rms current is 3.25A and the output is 12.1A. The design specifications are listed in Table I, and the sample parameters are listed in Table II.

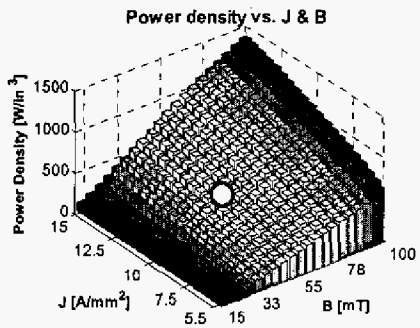
With the commercial magnetic core material, 3F4, and the in-house dielectric material, N1250, a design map was developed to demonstrate all of the feasible designs, as shown in Fig.13 (a). As the design variable, the average current density, J, varies in the range from 5 A/mm^2 to 15 A/mm^2, and the flux density, B, varies in the range of from 10 mT to 100 mT. By setting the copper thickness at 60% of the skin depth, the total loss was calculated. The loss map is shown in Fig. 13 (b). The picture of the sample and the small signal test result are shown in Fig. 14 and 15, respectively.

TABLE I
DESIGN SPECIFICATIONS

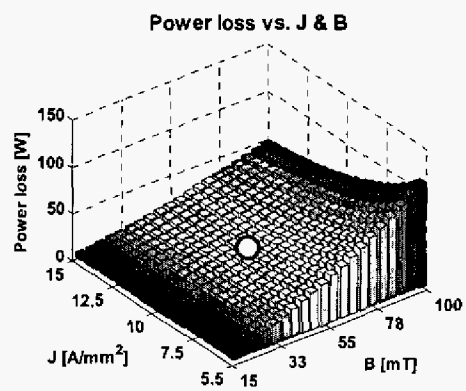
Parameters	Designed value	Real value
Resonant Inductance, L_r	0.4 uH	444 nH
Resonant Capacitance, C_r	50 nF	47.5 nF
Magnetizing Inductance, L_M	20 uH	21 uH
Turn's Ratio, $n_p : n_s : n_s$	4:1:1	4:1:1
Resonant Frequency, f_r	1.1 MHz	1.1 MHz
Operating Frequency, f	1 MHz	1 MHz

TABLE II
SAMPLE DIMENSIONS

Core material	3F4
Dielectric material	N1250
Leakage layer material	FPC C351
Core width	2 mm
Cell width	10 mm
Cell length	60 mm
Winding window height	4 mm
Copper thickness	40 μm
Power density	524 W/in^3

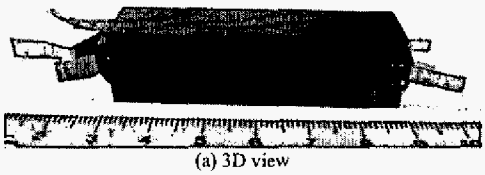


(a) Design map for power density

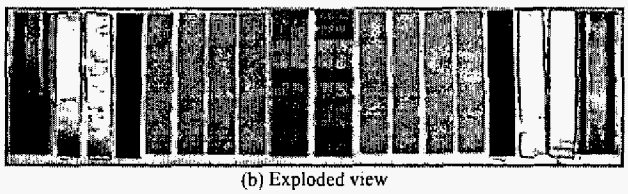


(b) Design map for power loss

Fig. 13 Design map for the LLCT module applied in a 500W converter



(a) 3D view



(b) Exploded view

Fig. 14 Integrated LLCT module

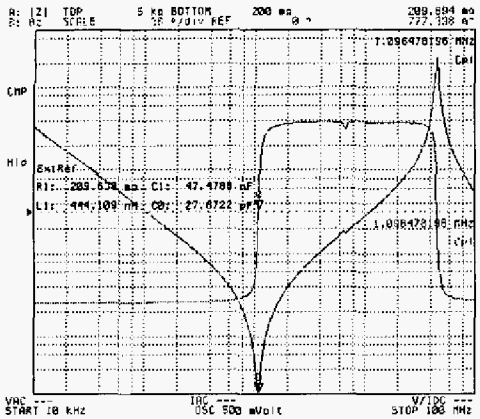


Fig. 15 Small signal test result of the LLCT module

VI. CONCLUSIONS

This paper demonstrates the theory of integrating capacitors and inductors into one passive module. The stacked structure is introduced as an optional structure with low loss density and high power density.

Due to the constructional characteristics of the stacked structure, a one-dimensional magnetic field can be assumed on the cross-section of the stacked structure. Both the electromagnetic design and loss calculation are based on this assumption.

As derived from the passive integration theory, the conduction current distribution in the primary winding is a linear function along the winding length. Correspondingly, the magnetic field distribution is also a function of the horizontal distance.

Based on the orthogonal principle of skin effect and proximity effect, the skin-effect loss and proximity-effect loss in the windings can be calculated respectively. Calculation results show that in the applications of the LLC resonant converter, the first three harmonics contribute most of the conduction loss, and the loss induced by the fundamental component occupies 85% of the total conduction loss. In the stacked structure, the secondary winding contributes most of the conduction loss. The calculation results also indicate that there exists an optimal copper thickness to minimize the conduction loss. With the assumption of a one-dimensional magnetic field, increasing the copper thickness will result in a constant conduction loss.

To explore these ideas, two design variables are set to develop the design map, which illustrates the possibility of high power density in the stacked structure.

REFERENCES

[1] B. Yang, F. C. Lee, A. J. Zhang and G. Huang, "LLC Resonant Converter for Front End DC/DC Conversion," *Applied Power Electronics Conference and Exposition, APEC'02*, vol. 2, pp. 1102 –1112.

[2] B. Yang "Topology Investigation for Front End DC/DC Power Conversion for Distributed Power System," Ph.D. dissertation, Virginia Polytechnic Institute and State University, Blacksburg, 2003

[3] B. Yang, R. Chen and F. C. Lee " Integrated Magnetic for LLC Resonant Converter," *Applied Power Electronics Conference and Exposition, APEC'02*, vol. 1, pp. 346 – 351.

[4] R. Chen, J. T. Strydom and J. D. van Wyk, "Design of Planar Integrated Passive Module for Zero-Voltage-Switched Asymmetrical Half-Bridge PWM Converter," *Industry Applications, IEEE Transactions*, vol. 39, No. 6, pp. 1648-1655

[5] R. Chen, J. T. Strydom and J. D. van Wyk, "Second Order Approximation Lumped Parameter Model for Planar Integrated L-L-C-T Module," *Industry Applications Conference, IAS'02*, vol.4, pp. 2419-2424.

[6] J. T. Strydom and J. D. van Wyk, "Electromagnetic Design Optimization of Planar Integrated Power Passive Modules," *Power Electronics Specialists Conference, PESC'02*, vol. 2, pp. 573-578.

[7] W. Liu, J. D. van Wyk and W. G. Odendaal, "Design and Evaluation of Integrated Electromagnetic Power Passives with Vertical Surface Interconnections," *Applied Power Electronics Conference and Exposition, APEC'04*, vol. 2, pp. 958-963.

[8] L. Zhao, "Generalized Frequency Plane Model of Integrated Electromagnetic Power Passives," Ph.D. Dissertation, Virginia Polytechnic Institute and State University, 2004.

[9] L. Zhao, J. T. Strydom and J. D. van Wyk, "A Planar Multicell Structure for Advanced Integrated Reactive Power Modules," *Industry Applications, IEEE Trans.* vol. 39, No. 6, pp. 1656-1664.

[10] W. Liu, J. D. van Wyk and W. G. Odendaal, "High Density Integrated Electromagnetic Power Passives with Vertical Interconnect and Stacked Structure," *Power Electronics Specialist, PESC'03*, vol. 1, pp.442-447.

[11] Jie Li, Tarek Abdallah, and Charles R. Sullivan, "Improved Calculation of Core Loss with Nonsimusoidal Waveforms", *Industry Applications Conference, IAS'01*, vol. 4, pp:2203 – 2210..

[12] R. Chen, J. T. Strydom, L. Zhao and J. D. van Wyk, "Large Signal Dielectric Characterization Measurement for Integrated Passive Devices," *Applied Power Electronics Conference and Exposition, APEC'01*, vol. 2, pp. 1203-1209.

[13] J. A. Ferreira, "Improved Analytical Modeling of Conductive Losses in Magnetic Components," *Power Electronics, IEEE Trans.* vol. 9, No. 1, pp. 127-131

[14] J. A. Ferreira, "Electromagnetic Modeling of Power Electronic Converters," Kluwer Academic Publishers, MA USA, 1989